Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/14/2025
PI Name:	Narayanan, Ranga Ph.D.		
Project Title:	A Novel Way to Measure Interfacial Tension Using the Electrostatic Levitation Furnace (ELF)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/22/2017	**End Date:**	08/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	5	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2025 per M. Sansoucie/MSFC (Ed., 4/23/25) NOTE: End date changed to 08/31/2024 per NSSC information (Ed., 6/26/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	A novel method using an Electrostatic Levitator (EL) is proposed to determine the surface tension of liquid/gas systems. This key thermophysical property is key to predicting optimum conditions for additive manufacturing, materials processing, and welding operations in space and on ground. The novel method for measurement of surface tension proposed involves the use of electrostatically forced Faraday instability. In this method, an oscillatory electrostatic forcing is applied to a liquid system, resulting in sudden pattern formation at the interface, i.e., an instability is caused. The "instability" is a result of resonance between the imposed forcing frequency and the natural frequency of the system. The critical amplitude and mode shapes are predictable from theory and compared with experiments. The measurement of surface tension directly aligns with the goals of the Thermophysical Property Measurement emphasis of the current NRA (NASA Research Announcement). Faraday experiments by this team using mechanical forcing have shown excellent agreement between theory and experimental results. However, in the case of mechanical forcing, the force of gravity overwhelms the effects of surface tension on the system stability and therefore it is almost impossible to determine this property with accuracy. Electrostatic forcing counters gravity, allowing the effects of surface tension to play a significant role in the ultimate determination of the system stability. Accordingly, experimentally mapped stability curves can be fit accurately with theoretical curves, allowing for the determination of the surface tension with great accuracy. The proposing team has successfully used electrostatic forcing in ground equipment to show that Faraday waves can be obtained as per theoretical predictions. The current state of the art uses an open container geometry to hold the liquid. However, a levitated sphere may also be used, and our proposal uses this containerless method. We therefore propose experiments with an important novel aspect. We propose to use ground-based facilities at JAXA (Japan Aerospace Exploration Agency) and NASA Marshall Space Flight Center (MSFC) to obtain tension for liquids using electrostatic Faraday forcing in using spherical drops by oscillating the drop over a large range of frequencies until the drop resonates with different modal patterns. The surface tension obtained from resonance at one modal patterns will be verified from deduced surface tension from another modal patterns. This will provide a self-consistent bench mark for determining surface tension. In summary, the research allows an alternate new method, viz., Faraday instability, for the determination of surface tension. Using electrostatic forcing instability will permit an independent confirmation of measurements obtained by the single-pulse oscillating-drop technique where decay rates are measured and related to surface tension and viscosity. The proposed method will also provide a slef consistent bench mark. Ground-based research using the electrostatic levitator available at the NASA MSFC and JAXA, and space experiments on ELF in the International Space Station (ISS) will provide accurate data. In addition to being an alternative method for surface tension measurement, the proposed technique also has versatility as a potential means in measuring surface tension for high viscosity materials and two-liquid interfacial tension on ground as well as in space. The proposing team has recently collaborated successfully with JAXA and has conducted experiments to study the feasibility of electrostatic Faraday instability. These results form the basis for the current proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Accurate surface tension values lead to transformative impact with space and Earth benefits such as a) Additive manufacturing of metals, where knowledge of surface tension is pivotal for predictive modeling to avoid rippling during solidification and possible wrinkling and shrinkage.. Human Exploration: Technology Area (TA) TA7.2.3.5: Welding, TA12.1.2: Computationally-Designed Materials, TA12.4.2.5: Additive Manufacturing. b) Space enabling operations. One example is joining operations such as soldering, where the surface plays an important role in the structure and stability of the weld. Accurate values for the surface tension will enhance the safety and effectiveness of this type of operation in space where the microstructures formed are different than those on Earth where gravity is present. c) Semiconductor crystal growth, where the interfacial tension between a liquid melt and liquid encapsulant plays a large role in determining conditions where convection is present.
Task Progress:	Over the past year, significant progress has been made in improving surface tension measurements of molten materials using electrostatic levitation and Faraday resonance. This contactless technique enables precise determination of thermophysical properties by eliminating contamination and container effects. The core methodology involves vibrating a levitated droplet and identifying its natural frequencies, which are directly linked to surface tension. Building on earlier work that focused on oscillation modes n=2 and n=3, this reporting period extended the analysis to over-forcing the first fundamental mode but getting responses for higher modes. By enhancing image processing and modal decomposition techniques, surface tension could be independently determined across multiple modal responses from a single trial. The approach uses projection into Legendre polynomial space to isolate each mode's contribution and quantify deformation amplitude with improved accuracy. Experiments conducted at NASA Marshall Space Flight Center (MSFC) using molten tin demonstrated that a single frequency sweep near the fundamental mode could excite several distinct modes as a response. The method was validated by comparing frequency peaks from both shape deformation and area change data, analyzed through Fourier transforms. The ability to identify and cross-check multiple modal responses provides internal consistency and increases measurement reliability. In summary, this year's work delivered a more robust and accurate method for containerless surface tension measurements. These advances enhance the potential for broader application across different materials and support future containerless processing efforts aligned with NASA's physical sciences objectives.
Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Dinesh B, Brosius N, Corbin T, Narayanan R. "Effect of a deep corrugated wall on the natural frequencies and the Faraday instability of a fluid interface." Phys. Rev. Fluids. 2024 Jul;9(7):073902. https://doi.org/10.1103/PhysRevFluids.9.073902 , Jul-2024

Articles in Peer-reviewed Journals	Ignatius I, Dinesh B, Dietze G, Narayanan R. "Thwarting Marangoni instability in viscoelastic liquid film via parametric forcing." Phys. Rev. Fluids. 2025 Apr;10(4):044001. https://doi.org/10.1103/PhysRevFluids.10.044001 , Apr-2025
Articles in Peer-reviewed Journals	Ignatius IB, Dinesh B, Dietze GF, Narayanan R. "Gravitational effects on Faraday instability in a viscoelastic liquid." J. Fluid Mech. 2025 May;1011:A28. http://dx.doi.org/10.1017/jfm.2025.328 , May-2025
Dissertations and Theses	Livesay J. "Experiments on resonance instability under variable gravity environments." Dissertation, University of Florida, May 2025. , May-2025
Patents	US Patent: US 12,135,269 B2 Patent issued November 5, 2024. Nov-2024 Brosius NB, Narayanan R, SanSoucie M. "System and Method For Measuring Surface Tension of a Levitated Sample."